

Introduction of the region

The study region focuses on a coastal boundary area along the Chilean coastline surrounding Valparaíso (Figure 1). The spatial extent of the region is bounded by longitudes 74.00° to 71.50°W and latitudes 35.50° to 33.00°S. Valparaíso is one of Chile's major coastal cities and serves as an important naval base, seaport, and fisheries hub (Fernández *et al.*, 2025). The Chilean coastline is influenced by the Humboldt Current, a cold ocean current that flows northward along the west coast of South America. This region forms part of the Eastern Boundary Upwelling System, which is commonly associated with coastal upwelling events that enhances marine productivity (Abrahams *et al.*, 2021; Fernández *et al.*, 2025).

This region was selected because of its significance as a major coastal and maritime area, as well as its close association with upwelling processes and high biological productivity (Trujillo *et al.*, 2018; Fernández *et al.*, 2025). In addition, the choice was made to focus on an area outside of the typically covered Benguela current.

Description of datasets and listings of metadata

Bathymetry data description

The bathymetry data consists of altitude data depicting the seafloor and land topography. Visualising the bathymetry of a region assists in understanding the shape of the continental shelf, continental slope and the deep-sea ocean floor.

The dataset was downloaded from the Global Multi-Resolution Topography Data Synthesis (GMRT) website (Ryan *et al.*, 2009). The dataset contains altitude data measured in meters. The dataset was specifically extracted from the boundary region mentioned earlier and is in NetCDF format. The negative values indicate the ocean depth, and the positive values indicate the land. The coordinate system for the dataset is according to the latitude and longitude coordinate system.

Listings of the metadata:

<xarray.Dataset> Size: 13MB

Dimensions: (lat: 1379, lon: 1141)

Coordinates:

- lat (lat) float64 11kB -35.5 -35.5 -35.5 -35.5 ... -33.0 -33.0 -33.0
- lon (lon) float64 9kB -74.0 -74.0 -74.0 -74.0 ... -71.5 -71.5 -71.5

Data variables:

- altitude (lat, lon) float64 13MB nan nan nan nan ... 164.0 163.0 176.0

Attributes:

- title: GMRT Grid
- history: Projection: Cylindrical Equidistant\nExtracted from the Glo...
- Conventions: COARDS,CF-1.6
- GMT_version: 4.5.7

Chlorophyll-a data description

The chlorophyll dataset used in this study is a global monthly climatology product from the ESA-CCI Ocean Colour project. Chlorophyll concentration is widely used as an indicator of phytoplankton biomass and therefore serves as an indication for biological productivity in the ocean.

The data are stored in NetCDF format and follow the Climate and Forecast (CF) metadata conventions, which ensure standardised structure and metadata for geoscientific datasets. The dataset consists of 12 monthly fields, covering the period from January 1998 to January 2020. Spatially, the data are organised on a regular latitude–longitude grid using a geographic coordinate reference system.

Listings of the metadata:

<xarray.Dataset> Size: 2GB

Dimensions: (time: 12, lat: 4320, lon: 8640)

Coordinates:

- time (time) datetime64[ns] 96B 1998-01-01 1998-02-01 ... 1997-12-01
- lat (lat) float64 35kB 89.98 89.94 89.9 89.85 ... -89.9 -89.94 -89.98
- lon (lon) float64 69kB -180.0 -179.9 -179.9 ... 179.9 179.9 180.0

Data variables:

- crs int32 4B ...
- chlor_a (time, lat, lon) float32 2GB ...

Attributes: (8/53)

- CDI: Climate Data Interface version ?? (htt...
- history: Tue Apr 27 20:27:21 2021: cdo selvar,c...
- source: NASA SeaWiFS, MODIS-Aqua, MERIS, VIIRS, and OLCI observations
- institution: Plymouth Marine Laboratory
- Conventions: CF-1.7
- Metadata_Conventions: Unidata Dataset Discovery v1.0
- time_coverage_start: 199801010000Z
- time_coverage_end: 202001312359Z

Figures and brief interpretations of the study region

Results interpretation

According to the plots, it's evident that the shelf of Valparaíso off Chile is quite narrow and then breaks with a steep gradient into the deeper waters. The shelf is deep relative to shelves off other coastal regions reaching 200 m. And deeper waters go up to 5000 m below the surface.

It was interesting to observe that the relatively higher chlorophyll-a concentrated waters were restricted inshore (Figure 2). The maximum is approximately 1.6 mg.m^{-3} particularly in two separate cells north and south of the coastline of Chile. This just clearly shows that over an annual basis particularly (September 1997-August 1998) the chlorophyll-a concentrations were higher in these two regions compared to adjacent areas. This indicates a possibility of importance for primary production also suggesting that nutrient availability might be higher in these areas along the coastline of Chile.

Monthly variations in chlorophyll-a concentrations were obvious (Figure 3), where larger areas of chlorophyll-a primary production were observed in the austral spring

and summer months, from September 1997 to February 1998, where chlorophyll concentrations extended over the narrow continental shelf. According to Aravena *et al.* (2024), this is not uncommon for this region. This area of primary production then narrows down onto the shelf directly for the austral winter months from March 1998 to August 1998. This is typical of systems that form part of eastern boundary current as they would typically bring nutrient rich waters to the surface from deeper offshore waters through physical processes such as localised upwelling events in summer.

Regarding the comparison between a single point of high chlorophyll-a concentration within this region, it was found that from September to December 1997, the regional chlorophyll-a concentration increased slightly but then decreased all the way through to July 1998 only for it to increase once more as we move to August of the same year showing some sort of continuity in terms of the seasonal repetition of trends. For the same temporal scale, the mean chlorophyll-a concentration from September to November 1997 saw a drastic increase, then immediately dropped to below 1.5 mg.mm⁻³ from then to May 1998. After this change was an increase from August.

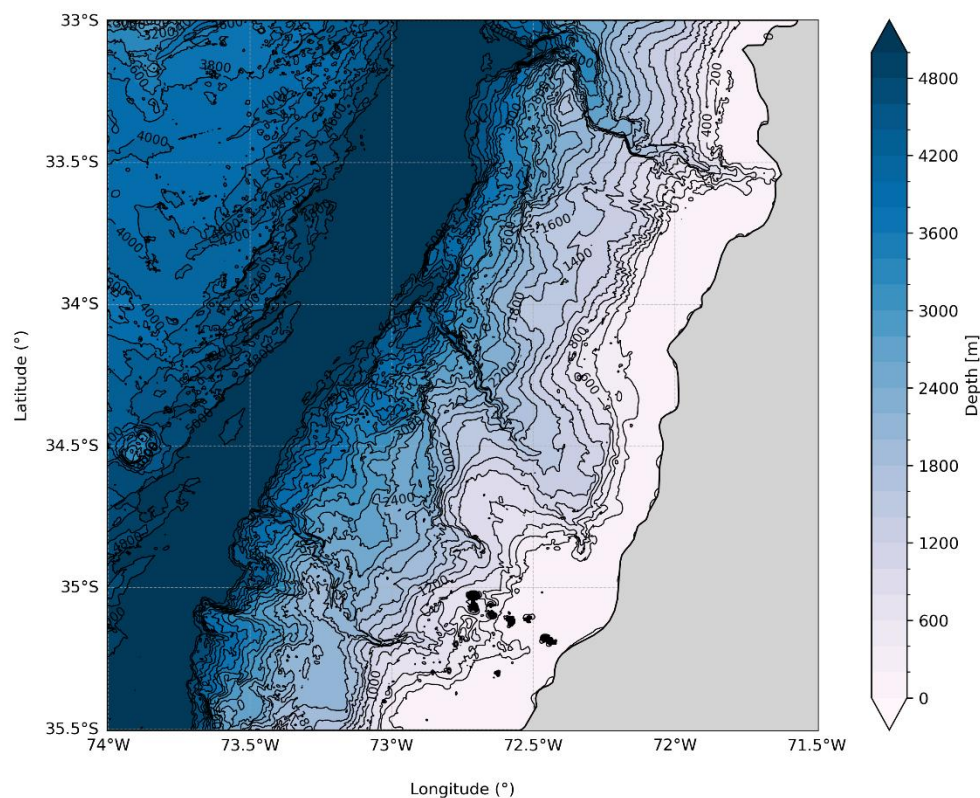


Figure 1: Bathymetric features associated with the coast of Valparaíso, Chile.

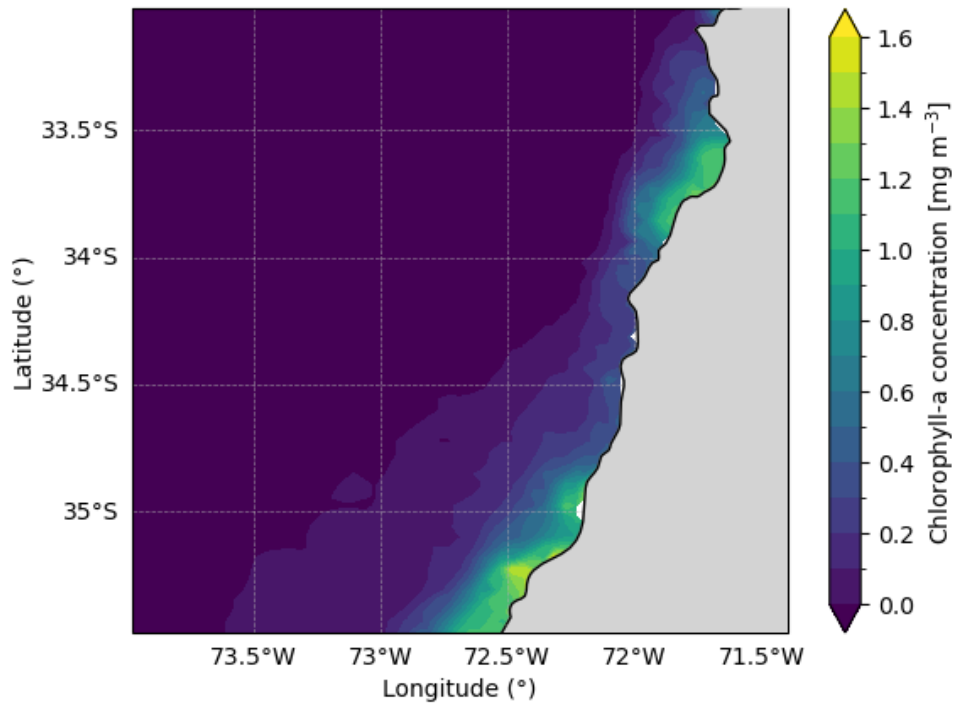


Figure 2: Log transformed chlorophyll-a concentration annual (1997-98) averages along the Chilean coastline surrounding Valparaíso.

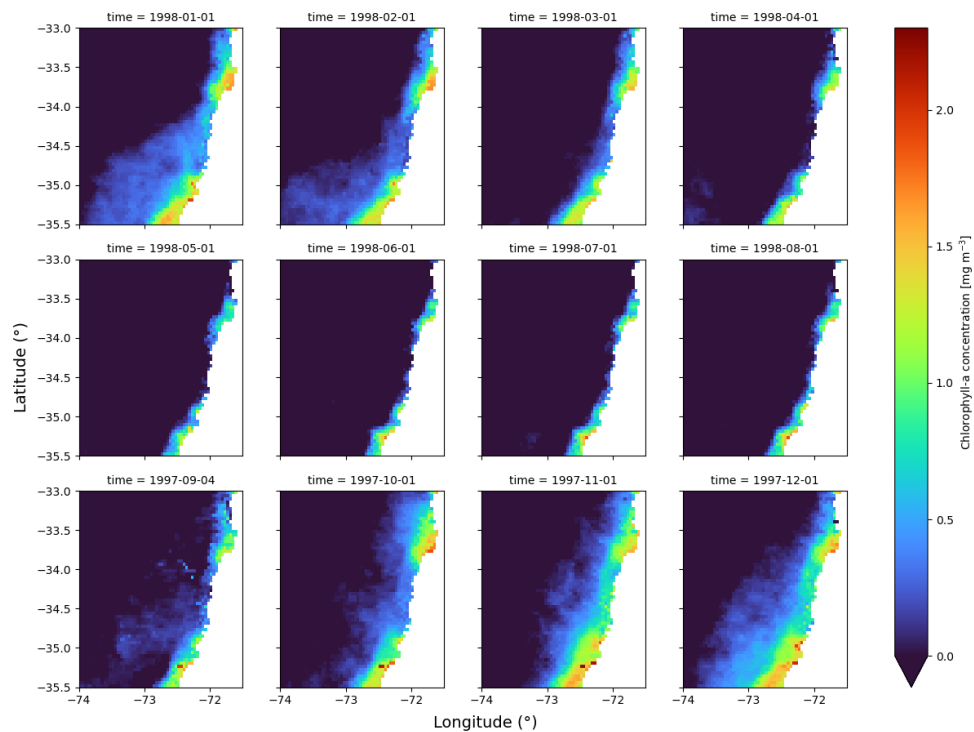


Figure 3: Climatology illustrating the log transformed mean chlorophyll-a monthly concentrations ($\text{mg} \cdot \text{mm}^{-3}$) between September 1997 and August 1998.

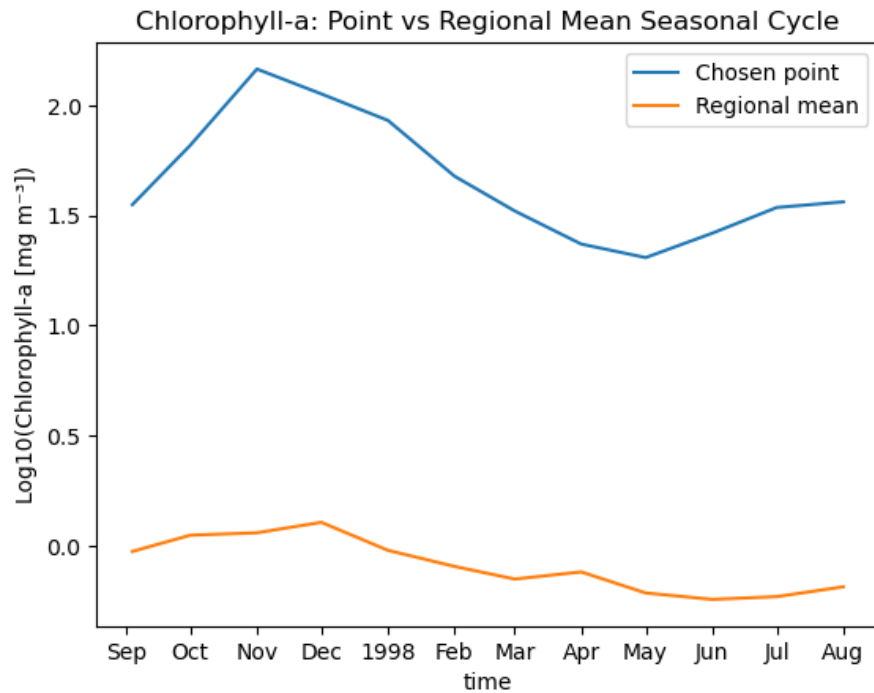


Figure 4: Comparing the log transformed mean chlorophyll-a concentration annual time-series of the greater Chile coastline region to the single point of high concentrations.

Disclaimer of AI use

Artificial Intelligence (AI) was used to achieve some of the goals set out in the assignment. The use of AI was mainly used during the coding section of this assignment. A large model language such as ChatGPT was used to assist with fixing errors in pre-written code. There were no issues plotting the bathymetry data; however, some issues arose regarding the appearance of the plot. I preferred having the altitude data as Depth (m) and that required me to convert the negative variables and contour labels into positive values. An example of my prompt would include the code itself followed by “Why is this code `depth = ds_bath.altitude * -1` not working”. The response would include the corrected code such as “`-ds_bath.altitude`”. Other issues occurred such as the layering of the projections. An prompt example would include the code and “Why is the `ax.add_feature` not displaying the coastline of land feature”. The response indicated that the code which plots the gridlines overrides the order of the code and forcibly plots it at the top and introduced function `zorder` which manually orders the layers. This function was useful for later applications. The gridline feature also interrupted labelling the axis and therefore the prompt also included “Why is the code to plot the axis labels not working”. This was similar to the previous case, and

the response was to expand on the code with `ax.text(0.5, -0.09, 'Longitude (°)', transform=ax.transAxes, ha='center')`.

References

- Abrahams, A., Schlegel, R.W. & Smit, A.J. 2021. Variation and change of upwelling dynamics detected in the world's eastern boundary upwelling systems. *Frontiers in Marine Science*, 8 : 626411.
- Aravena, G., Broitman, B. & Stenseth, N.C. 2014. Twelve years of change in coastal upwelling along the central-northern coast of Chile: spatially heterogeneous responses to climatic variability. *PLoS One*, 28; 9 (2) : e90276. doi: 10.1371/journal.pone.0090276. PMID: 24587310; PMCID: PMC3938675.
- Fernández, M., Aldana, M., Duarte, C., García-Huidobro, M.R., Varas, O., Estévez, R.A., Pulgar, J., Quijón, P.A. & Pulgar, J. 2025. Exploring the relationship between upwelling intensity and socio-ecological attributes of marine exploitation areas for benthic resources (MEABRs), along the southern Humboldt Current system. *Journal of Environmental Management*, 374 : 124102.
- OpenAI. 2026. ChatGPT (May 17 version) [Large language model]. Available at: <https://chatgpt.com/> [Accessed 15 May 2026].
- Ryan, W.B.F., S.M. Carbotte, J.O. Coplan, S. O'Hara, A. Melkonian, R. Arko, R.A. Weissel, V. Ferrini, A. Goodwillie, F. Nitsche, J. Bonczkowski. & R. Zemsky. 2009. Global Multi-Resolution Topography synthesis, *Geochem. Geophys. Geosyst.*, 10: Q03014. doi: [10.1029/2008GC002332](https://doi.org/10.1029/2008GC002332)
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